

Nithya Madalakote Prakash

AI/ML Engineer · Generative AI & RAG Systems · Berlin, Germany

nithyamadalakotepakash12@gmail.com

linkedin/nithya-prakash

github/nithya-prakash

portfolio/nithya-prakash



PROFESSIONAL SUMMARY

AI/ML Engineer building evaluation-driven Generative AI systems with measurable outcomes: hybrid RAG (0.986 NDCG@5), multimodal diagnostics (271-test suite), industrial IoT (385 msg/s), and predictive maintenance (3.06-cycle RMSE). Strong software engineering foundation in Python, FastAPI, and Kubernetes, with expertise in rigorous evaluation frameworks, hybrid retrieval architectures, and production-oriented ML pipelines.

TECHNICAL SKILLS

Generative AI & RAG: LLMs · RAG Pipelines · Hybrid Retrieval (Dense + BM25 + RRF) · Cross-Encoder Reranking · Prompt Engineering · Hallucination Detection · LLM Evaluation (Recall@K, MRR, NDCG@K)

Machine Learning & Evaluation: PyTorch · Scikit-learn · SHAP · Optuna · MLflow · Anomaly Detection

Backend & Infrastructure: Python · FastAPI · PostgreSQL · Docker · Kubernetes · Kafka

Software Engineering: Git · pytest · REST APIs

PROJECTS

HybridRAG (RAG Knowledge Assistant) Python · FastAPI · Qdrant · Ollama · Next.js · PostgreSQL

[github](#)

- Evaluated hybrid RAG (dense + BM25 + RRF + cross-encoder reranking) across 110 labeled queries: 100% Recall@5, 0.980 MRR, 0.986 NDCG@5; benchmarked 5 retrieval configurations with ~1.08s reranking p50 latency.
- Designed answer-decline guard achieving 95.5% accuracy and 0.762 F1 across 110 queries; root-caused 72.7% recall gap to reranker overconfidence on topically similar but incorrect content.

Industrial Multimodal AI Copilot Python · Qdrant · Celery · FastAPI · Docker · PostgreSQL

[github](#)

- Built hybrid retrieval pipeline (dense + BM25 + RRF + cross-encoder reranking); used component ablations to identify and fix silent dense-retrieval bug, improving MRR to 0.886 vs. 0.878 BM25-only on 7-query benchmark.
- Maintained 271-test suite (real Postgres/Qdrant/Docker integration, no infra mocks); found and fixed worker concurrency bug silently breaking every second document-processing job.

InduSense — Industrial IoT Monitoring Platform Python · FastAPI · Kafka · MQTT · Kubernetes · InfluxDB

[github](#)

- Built event-driven industrial IoT platform processing telemetry from 1,000 simulated sensors via MQTT→Kafka→stream processing→anomaly detection→alerting; accounted for all 127,374/127,374 telemetry messages with rule-based, statistical, and Isolation Forest detection.
- Engineered Dockerized, Kubernetes/Helm-based infrastructure with Prometheus/Grafana; verified 385 msg/s throughput, 74.3 ms mean alert latency, and 146-test CI suite (139 passed, 7 skipped).

Manufacturing Predictive Maintenance AI Python · Scikit-learn · SHAP · FastAPI · MLflow · Optuna

[github](#)

- Architected predictive-maintenance pipeline for RUL and failure prediction, achieving 3.06-cycle RMSE in critical ≤30-cycle range and 9.72-cycle overall RMSE on machine-held-out test set.
- Engineered end-to-end FastAPI ML pipeline with time-series feature engineering and SHAP explainability; benchmarked 68.21 ms p99 latency across 100 sequential local requests.

WORK EXPERIENCE

- Java Full Stack Developer Trainee · **Palle Technologies**

July 2024 — Jan 2025, Bengaluru, India
- Delivered 8+ production features across Java backend and React frontend; built 150+ unit tests achieving 87% code coverage; resolved 25+ defects through systematic root-cause analysis.
 - Optimized 12+ SQL queries reducing API latency by 40% on high-traffic endpoints; maintained Git workflows and deployed features across staging environments.
 - Participated in code reviews and Agile ceremonies; documented technical decisions and contributed to team knowledge base.

EDUCATION

- M.Sc. in Artificial Intelligence · **Brandenburgische Technische Universität**

Oct 2025 – Present · Germany
- Neural Signal Analysis · Image Processing & Computer Vision · Causal Data Science · Modeling of Perception and Action
- B.Tech. in Computer Science · **Presidency University**

2020 – 2024 · Bengaluru, India
- GPA: 7.62 / 10 (equivalent grading scale)

PUBLICATIONS

- MediSync: Streaming Healthcare Continuity — IJRPR (ISSN 2582 - 7421)

2023

LANGUAGES & CERTIFICATES

Languages

- English — C1 (Fluent)
- German — A2 (currently pursuing)

Certificates

- Introduction to Machine Learning - Kaggle
- The Complete Data Science Bootcamp - Udemy